

Jacob Lamadrid

Data Science | Machine Learning | Operations Research

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EDUCATION

•Master of Science in Analytics

Georgia Institute of Technology

Atlanta, GA

Expected: Dec 2027

- Focus: Modeling, Simulation, and Optimization

•Bachelor of Science in Cognitive Science - Machine Learning and Neural Computation

University of California, San Diego

San Diego, CA

Jun 2024

- Minor: Data Science
- Provost Honors | 2024 Cognitive Science Summer Scholar (\$6,000 Award)
- Instructional Assistant: Introduction to Python, Introduction to Data Science

EXPERIENCE

•Graduate Data Science Intern

Centene Corporation

May 2026 – Aug 2026

St. Louis, MO

- Developed end-to-end data pipelines unifying internal audit and claims warehouse sources into a common schema of 127k+ claims, enabling supervised anomaly detection on high dimensional, highly imbalanced (50:1) data
- Applied Bayesian Additive Regression Trees (BART) to quantify predictive uncertainty on flagged records, enabling risk-tiered prioritization that focused reviewer effort on high-confidence anomalies and reduced false-positive volume 10x

•Research Fellow

UC San Diego Natural Computation Lab

Jul 2024 – Sep 2024

San Diego, CA

- Conducted independent research on signal reconstruction from EEG data using CNNs, Latent Diffusion Models, and DCGANs; managed full ML lifecycle from multimodal data ingestion to model evaluation
- Designed signal processing pipelines for EEG and audio data; translated theoretical research into reproducible PyTorch experiments on custom neuroimaging datasets

•Software Development Intern

NASA Langley Research Center

Jun 2023 – Sep 2023

Hampton, VA

- Developed end-to-end data pipelines in Python to standardize and migrate legacy atmospheric science datasets to the NASA TOLNet repository, ensuring format compliance across multiple research instruments and sites
- Built automated data quality analytics to flag anomalies and inconsistencies in historical lidar datasets; delivered summary reports and documentation to cross-functional research teams

•Application Developer & Research Data Associate

UC San Diego Health – Autism Center of Excellence

Jun 2022 – Sep 2024

San Diego, CA

- Built TrackiT!, a cross-platform clinical app (Flutter/Dart/Firestore) for longitudinal tracking of developmental trends across neurodevelopmental studies
- Designed GUI tools and algorithmic pipelines for clinical data entry, eye-tracking QA, and behavioral assessments, supporting downstream statistical modeling workflows

PROJECTS

•Bayesian Temporal Signal Extraction & Uncertainty Quantification

R, RStan, RShiny, Variational Inference, Sensor Array Deconvolution

2025

- Engineered a Bayesian deconvolution pipeline for sparse high-dimensional sensor arrays; designed automated QC using posterior uncertainty estimates to separate latent signals from stochastic noise and artifacts

•Meteor Instrument – Embedded Radio Astronomy & Meteorology System

Python, Raspberry Pi, C++, Arduino, Embedded Systems, Signal Processing

Ongoing

- Developing a standalone embedded instrument on Raspberry Pi and Arduino display hardware for real-time radio astronomy and meteorology data capture, processing, and visualization

•WaveMAP – Neural Waveform Clustering & Analysis

Python, Jupyter, Clustering, Dimensionality Reduction, Waveform Feature Extraction

2024

- Applied WaveMAP-style clustering to high-dimensional neocortical waveform recordings; built a reproducible feature-extraction and dimensionality-reduction pipeline used as a QA tool for identifying true signal events

TECHNICAL SKILLS

Languages: Python, R, MATLAB, Julia, SQL, C++, JavaScript, Dart

Frameworks & Tools: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, RStan, RShiny, Spark, Firestore, Git